sciendo

Management Systems in Production Engineering

2024, Volume 32, Issue 4, pp. 571-586

Date of submission of the article to the Editor: 01/2024
Date of acceptance of the article by the Editor: 10/2024

DOI 10.2478/mspe-2024-0055

BLOCKCHAIN IN ENTERPRISE RESOURCE PLANNING SYSTEMS: A COMPREHENSIVE REVIEW OF EMERGING TRENDS, CHALLENGES, AND FUTURE PERSPECTIVES

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Abstract:
This research paper conducts a systematic review of the integration of blockchain technology with Enterprise Resource Planning (ERP) systems, a burgeoning field at the intersection of advanced technology and business management. The primary objective is to bridge the knowledge gap by synthesizing existing literature, providing insights into the current state, challenges, and potential of blockchain in enhancing ERP systems. The study follows a structured methodology involving four key stages: Literature Retrieval, Literature Screening, Bibliometric Analysis, and Content Analysis. A comprehensive literature search in the Scopus database resulted in the identification and analysis of 250 scholarly articles, narrowed down to 66 relevant publications. The findings highlight a significant trend towards decentralized ERP solutions, driven by blockchain's promise of enhanced security, transparency, and efficiency. Key areas impacted by blockchain integration include supply chain management, accounting, security, traceability, purchasing, and process automation within ERP systems. The bibliometric analysis reveals a growing academic interest in this domain, with a focus on scalability, data security, and interoperability. However, the integration faces challenges such as technical complexity, cultural resistance, and regulatory uncertainties. The study concludes that while blockchain presents transformative opportunities for ERP systems, realizing this potential requires overcoming substantial technical and organizational hurdles. Future research should concentrate on developing scalable, interoperable blockchain solutions, tailored to specific industry needs and compliant with regulatory standards, thereby facilitating widespread adoption and maximizing the benefits of blockchain in ERP systems.

Key words: blockchain technology, enterprise resource planning (erp), supply chain management, process automation, data security; smart contracts

INTRODUCTION

Enterprise Resource Planning (ERP) systems are integrated software solutions designed to manage and streamline all core business processes. These systems facilitate the unification of various business functions within an organization, ensuring seamless data flow and operational efficiency. Typically, ERP systems encompass a range of modules, each tailored to specific aspects of business management. These modules usually feature a variety of key areas, such as financial and cost accounting, sales and distribution, materials management, human resources, and production planning, among others [1].

Advancements in technology and evolving business practices have driven substantial evolution of ERP systems. The emergence of internet and web technologies in the early 2000s instigated a shift towards more open and flexible ERP architectures, often termed as ERP II [1]. More recently, the advent of cloud computing has made ERP systems more scalable, flexible, and cost-efficient, broadening the range of businesses that can utilize ERP functionality [2]. The incorporation of data analytics, artificial intelligence, and machine learning into ERP systems has improved decision-making and predictive capabilities, amplifying their value to contemporary businesses [3].

Blockchain is a distributed ledger technology that records transactions in a secure, transparent, and tamper-evident manner. The process begins with the submission of a transaction (TRX), which is then broadcast to a Peer-to-Peer (P2P) network. Nodes within this network run a consensus protocol to validate the TRX, ensuring its legitimacy and compliance with the network's rules. Once validated, the TRX is combined with other transactions to form a new block, which is then appended to the blockchain, creating an indelible link to the previous block. This sequential linking of blocks forms a chain, and each block is immutable once added, making the ledger a trusted and

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verifiable record of all transactions from inception to the present. The ability to write new transactions to the blockchain can be open to all or restricted to certain authenticated users, depending on whether the blockchain is public or private, as governed by the consensus protocol [4]. The blockchain technology, was originally designed as a public transaction ledger for Bitcoin cryptocurrency [5]. It functions as a decentralized and distributed digital ledger, recording transactions across multiple computers in a way that any recorded data cannot be retroactively altered without changing all subsequent blocks. Its potential to revolutionize various sectors beyond cryptocurrencies has been recognized since its inception. As a potentially transformative force, blockchain technology promises decentralization, transparency, and immutability in sectors like healthcare, supply chain management, and finance [6]. For instance, in supply chain management, blockchain can improve traceability, minimize fraud, and enhance contract enforcement [7].

Blockchain's integration with ERP systems promises unparalleled security, transparency, and efficiency. It can enhance data integrity, streamline processes, and improve auditability [8]. However, despite the promising potential of blockchain in enhancing ERP systems, practical applications are still in a nascent stage. The integration of blockchain technology into ERP systems represents a promising area that has yet to be fully explored. Despite the potential of blockchain to enhance data security, transparency, and efficiency, there is a lack of comprehensive research synthesizing these scholarly efforts [8]. This research gap limits our understanding of the implications and benefits of blockchain integration with ERP systems, potentially slowing its broader adoption and hindering the realization of its full potential.

Despite the extensive body of research independently surrounding both ERP and blockchain technologies, the intersection of these two critical fields remains notably underexplored. While there are sporadic studies that touch upon the potential synergies and challenges of integrating blockchain into ERP systems, a noticeable lack exists in terms of a holistic and integrative review of the collective research in this sphere [9, 10, 11]. This paper endeavors to bridge this significant research gap by undertaking a thorough synthesis and critical analysis of the existing literature pertaining to the amalgamation of blockchain technology with ERP systems. The objective of this study is to offer a thorough and detailed insight into the present situation, identifying areas that have not been extensively studied and outlining a defined and organized direction for future academic research. This work is valued not just academically, but also holds substantial practical value. A detailed and panoramic view of the current state of research is being offered by this work, positioning it to serve as a valuable resource for both the academic community and industry practitioners, guiding future explorations and applications in this promising and dynamic domain.

This paper is structured as follows: Section 2 details the methodology, outlining the systematic process used for literature retrieval, screening, bibliometric analysis, and content analysis. Section 3 delves into the bibliometric analysis, offering insights into research trends, significant contributors, and thematic clusters in the field. Section 4 encompasses a detailed content analysis, examining how blockchain integration impacts various facets of ERP systems. The paper ends in Section 5 with a Conclusion, where we succinctly summarize the key findings and their broader implications for the future landscape of ERP and blockchain integration. This section also synthesizes the insights, outlines prospective research avenues, and critically examines the limitations of current studies in this evolving field.

METHODOLOGY

The methodology used in this research adheres to a structured and systematic process, meticulously designed to collect and analyze literature pertaining to the integration of blockchain technology with ERP systems. This process is broken down into four essential stages, each playing a crucial role in ensuring a comprehensive understanding of the topic. These stages are outlined as follows.

Literature Retrieval

Our research methodology employed a comprehensive and targeted data collection strategy, leveraging the Scopus database. In this study, we selected the Scopus database for our literature review due to its comprehensive coverage and extensive range of peer-reviewed journals. Scopus is recognized for its broad scope across various disciplines, including both well-established and emerging fields, and offers robust search functionalities. This choice ensured a thorough and inclusive analysis, aligning with our aim to capture the most relevant and up-to-date research findings. We meticulously formulated our search criteria using a combination of specific keywords: "Enterprise Resource Planning Systems" or "ERP", each paired with terms critical to our study, including “Blockchain”, "Decentralization", "Cryptocurrency", "Smart payment", "Smart contract", "Smart report", "Cryptography". This strategic approach extended beyond mere keyword matching; we conducted an in-depth search that encompassed titles, abstracts, and keyword fields to unearth potential connections between ERP and blockchain systems. This thorough search process concluded in the identification and enlistment of an extensive collection of 497 scholarly papers, covering a significant time span from 2018 to 2023. This rich dataset forms the foundation of our research, offering a diverse range of perspectives and insights into the integration of blockchain technology with ERP systems.

Literature screening process

Adhering to the principles of the PRISMA statement, a widely recognized framework for systematic reviews and meta-analyses, a systematic literature screening process was implemented. This facilitated the methodical identification, selection, and evaluation of relevant literature. Initially, 497 papers were retrieved. Upon the elimination of duplicates, this number was reduced to 305. Each

publication was then subjected to a thorough examination, with those deemed irrelevant being excluded (Figure 1). This comprehensive screening process resulted in a refined list of 66 pertinent reviews, papers, and publications, all falling within the timeframe of 2018 to 2023.

The focus of the study was on exploring blockchain platforms and their potential for enhancing the development of structured ERP systems. A diverse array of sources was consulted, encompassing books, journal articles, and conference papers. Figure 2 illustrates the growth in the volume of blockchain-related papers over the past five years, effectively showcasing the progression in the field's research during this period. It should be noted that the data collection concluded at the end of June 2023, which accounts for the comparatively lower count of papers in 2023.

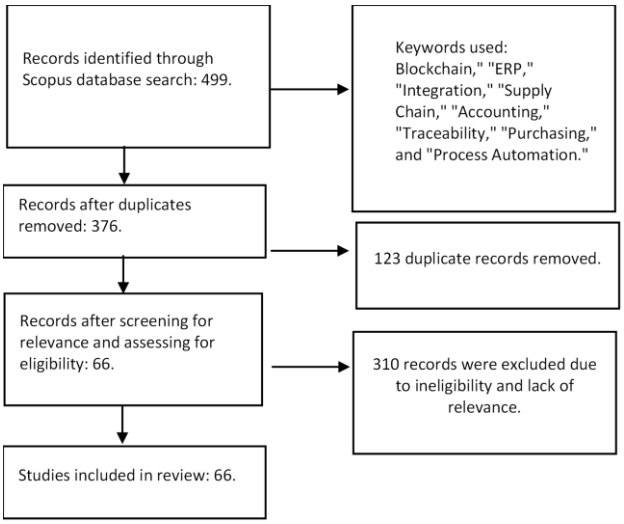


Fig. 1 Literature screening approach

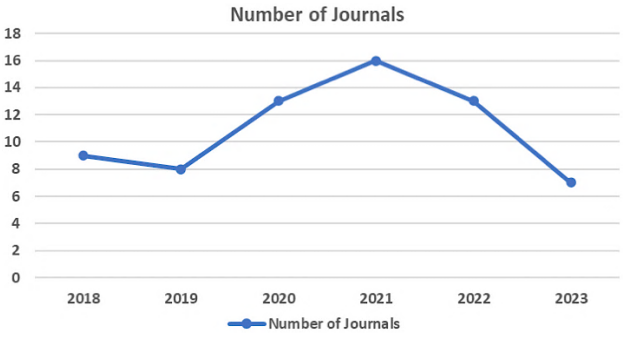


Fig. 2 Number of publications in relation to blockchain and ERP systems integration (2018-2023)

The results show that the phrase "Blockchain with correlation to ERP system" first appeared in the scientific literature in 2018 and that there has since been a steady increase in the number of publications. This is evident in the figure with reference to the impact of COVID-19 on the use of technology, where we can see a huge interest in the topic post-pandemic due maybe to the heavily rely on technology to manage our lives, work, and businesses in general.

Bibliometric Analysis

Employed as a methodical assessment tool for academic literature, bibliometric analysis focuses on the detailed scrutiny of citations and references within scholarly papers. This technique delves into the examination of citation patterns, co-authorship relations, and keyword frequencies. Through such analysis, it aids in identifying seminal works, pinpointing key contributors in the field, uncovering nascent areas of research, and mapping out the collaborative networks among scholars. Significantly, bibliometric analysis offers critical insights into the developmental trajectory of research themes, thereby facilitating the recognition of prominent researchers and prevailing trends in the intersecting domain of blockchain and ERP integration.

Content Analysis

During this phase, a thorough examination and systematic organization of the gathered research publications were conducted. Utilizing content analysis, the data was categorized and classified into distinct themes and subthemes. This methodical approach enabled the identification of prevailing patterns and common threads across the collected data. Specifically, in the context of this study, the content analysis was tailored to illuminate the numerous ways in which the integration of blockchain technology with ERP systems impacts various facets of enterprise operations. This detailed examination provided a deeper understanding of the relationship between blockchain and ERP systems, highlighting key areas of influence and potential implications for enterprise management and strategy.

BIBLIOMETRIC ANALYSIS

Despite the widespread acknowledgment of the demand for blockchain and ERP technologies across various industries, there is a notable scarcity of actual implementations of blockchain within ERP systems. Existing literature primarily focuses on the intersection of ERP and blockchain, discussing the benefits and challenges associated with their integration. However, a comprehensive understanding of their current application and future prospects remains limited. To bridge this knowledge gap, a bibliometric analysis was conducted, aiming to assess the extent of recognition and application of ERP and blockchain in different sectors over the past five years. This analysis utilized the final selection of 66 pertinent papers from the Scopus database, specifically focusing on ERP and blockchain. Tools such as VOSviewer and Microsoft Excel were employed in this phase to analyze the data meticulously, with the objective of identifying prevailing trends and generating visual mapping of these trends. This approach facilitated a more nuanced understanding of the current landscape and potential trajectories in the integration of ERP and blockchain technologies.

Four distinct types of visualization maps were generated using VOSviewer to conduct a comprehensive analysis of the current state of blockchain platforms within ERP systems. In these maps, each circle, or 'node', represents a

specific term. The size of each circle, along with the font size of the term, indicates the frequency of the term's usage in the literature: larger circles and bolder typeface correspond to terms that are more frequently mentioned. The spatial proximity of any two terms in the map reflects their thematic relatedness in the research field. Terms that are closely positioned are more strongly related, suggesting a higher degree of association or co-occurrence in the literature. This visualization technique provides an intuitive understanding of the key themes and their interconnections within the realm of blockchain in ERP systems.

Co-occurrence map based on text data

Through the analysis of text data extracted from journal articles, the study successfully identified frequently recurring terms along with their associated sub-terms. This method facilitated the detection of evolving patterns within the blockchain field, including the emergence of new and influential terms. A total of 354 terms were extracted for this analysis. The study considered all these terms, including those that appeared less frequently or not at all, to ensure a comprehensive understanding of the domain. This inclusive approach, as depicted in Figure 3, allows for a more nuanced exploration of the field, capturing both dominant and emerging concepts in blockchain research.

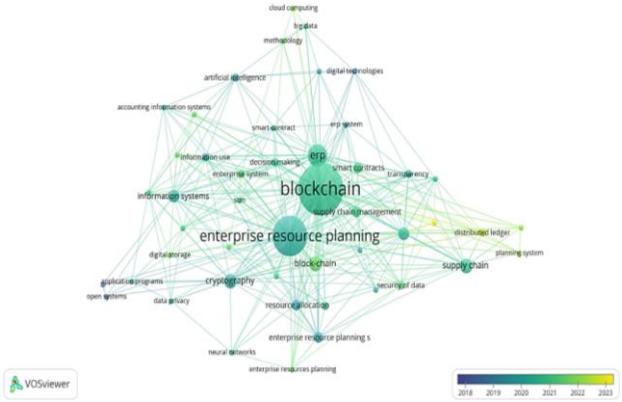


Fig. 3 Co-occurrence analysis

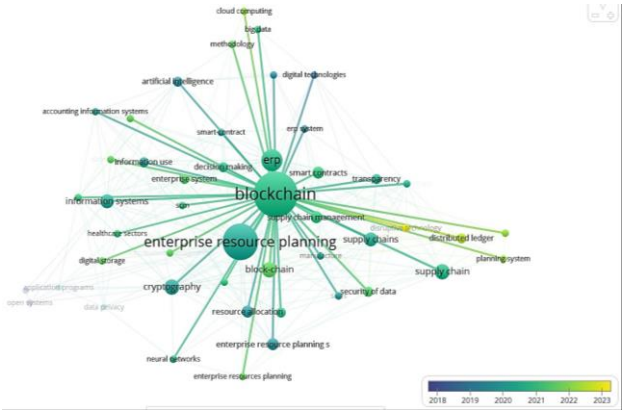


Fig. 4 Terms directly connected with Blockchain

Figure 4 offers a visual representation, highlighting the terms that are directly associated with blockchain. This visualization aids in understanding the contextual relationship between blockchain and its related concepts within the scope of the study.

Table 1 reveals insights into the frequency of certain terms within the field of blockchain and ERP research. 'Blockchain' leads with 29 occurrences, underscoring its central role and burgeoning interest in the field. Interestingly, 'Enterprise Resource Planning' and its abbreviation 'ERP' appear 21 and 10 times respectively, indicating significant but varying emphasis on ERP in the context of blockchain.

Table 1

Top 10 terms by occurrences

Rank	Term	Occurrences	Relevance Score
1	Blockchain	29	0.9209
2	Enterprise Resource Planning	21	0.8079
3	ERP	10	0.3927
4	Cryptography	6	0.2938
5	Block-chain	6	0.2655
6	Supply Chain	6	0.1469
7	Information Systems	5	0.2203
8	Supply Chains	5	0.1610
9	Supply chain Management	4	0.1525
10	Resource Allocation	4	0.1497

The appearance of 'Cryptography' (6 occurrences) alongside 'Blockchain' highlights the technological and security aspects that are pivotal in blockchain discussions. The term 'Supply Chain', also occurring 6 times, suggests a strong link between ERP systems and supply chain management, an observation further supported by the presence of 'Supply Chains' and 'Supply Chain Management' in the list. 'Information Systems', appearing 5 times, and 'Resource Allocation', with 4 occurrences, indicate the broadening scope of blockchain application, encompassing information system strategies and resource management. This data not only reflects the current research focus but also hints at the evolving trends in the integration of blockchain with ERP systems.

Co-occurrences map based on keywords

The keyword co-occurrence map (Figure 5) yielded seven distinct keywords that emerged in more than five instances, highlighting their prominence in the field.

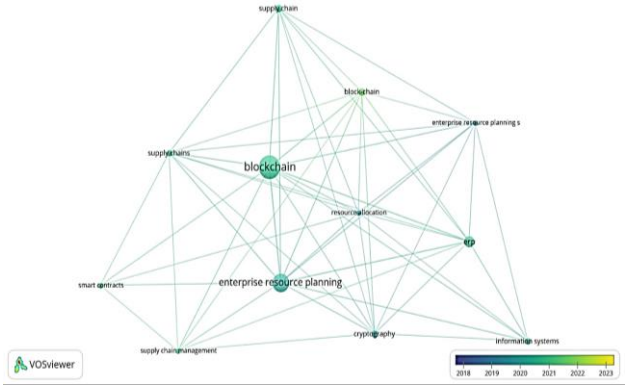


Fig. 5 Co-occurrence map of keywords

A notable trend in recent publications is the heightened emphasis on data security, reflecting the growing concerns and interests in this area. Among these recurrent keywords, 'cryptography' and 'information systems' stand out for their strong associations with blockchain technology. This analysis indicates a significant shift towards exploring the synergies between blockchain and ERP systems, with an increasing focus on how these technologies can be integrated with information systems. Particularly, the rising interest in smart contracts is indicative of a broader trend: the move towards a holistic approach that seamlessly melds blockchain, ERP systems, and information systems. This trend underscores a growing recognition of the potential for blockchain to enhance the security, efficiency, and overall functionality of ERP systems within the digital infrastructure of modern enterprises.

Co-occurrence map based on country of co-authorship

The data analysis also encompassed the geographical distribution of these publications. A visualization map was created to showcase the co-authorship country, considering a minimum requirement of 10 documents from each country. Out of the 25 countries, 12 met this threshold, and the results are depicted in Figure 6.

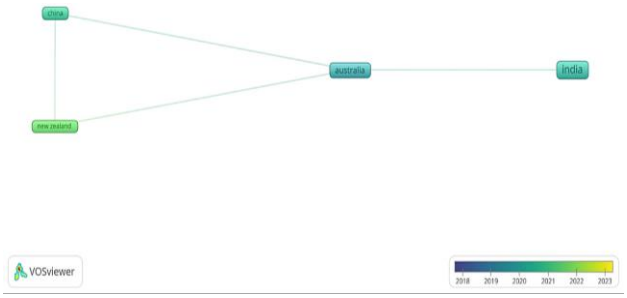


Fig. 6 Country of co-authorships

The majority of publications are associated with Australia, followed by India and China. Additionally, it was observed that there is a rising interest in Blockchain technology in countries such as New Zealand and various European Union (EU) countries like France, Belgium, Germany, Greece, among others.

Co-occurrence based on sources

The analysis presented in Figure 7 reveals a comprehensive overview of the journals dedicated to exploring the correlation between ERP and blockchain.

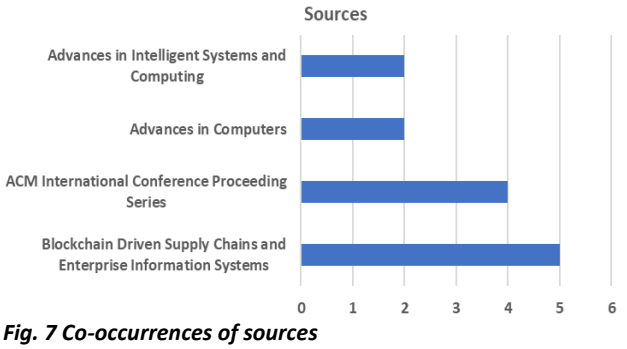


Fig. 7 Co-occurrences of sources

Notably, these journals originate from 52 distinct sources, demonstrating the widespread interest and engagement in this area of research. Among these publications, certain journals have emerged as significant contributors, with noteworthy representation. Specifically, "Blockchain Driven Supply Chains and Enterprise Information Systems" stands out with five publications, followed by the "ACM International Conference Proceeding Series" with four publications. Additionally, "Advances in Computers" and "Advances in Intelligent Systems and Computing" have each contributed two publications, underscoring their importance in the field. The diverse range of publishers and the varying number of publications highlight the depth and breadth of scholarly exploration into the integration of ERP and blockchain within these 66 selected papers.

CONTENT ANALYSIS

The content analysis section of this study methodically investigates the impact of blockchain technology on ERP systems, delineating the transition from centralized to decentralized models of data management. The section dissects the implications of this integration across diverse sectors, highlighting the potential for enhanced transparency, security, and efficiency. It synthesizes findings from key research studies, providing insights into how blockchain technology redefines supply chain management, accounting practices, and overall enterprise operations. This analysis concludes in a discussion of the challenges and opportunities presented by the blockchain-ERP convergence, setting the stage for further exploration into the transformative effects of this integration.

ERP & Decentralization

ERP systems aim to streamline and optimize business processes, providing a centralized framework for data and resource management. Decentralization, on the other hand, refers to the distribution of functions and powers away from a central, authoritative location or group. In the context of blockchain technology, decentralization is a foundational principle. Blockchain enables a distributed ledger system where data is stored across a network of computers, making it transparent, secure, and resistant to tampering. This decentralization aspect of blockchain is revolutionary for various industries, including ERP systems.

Manceski & Nechkoska [11], highlight the integration of blockchain into ERP systems as a strategy to develop a decentralized, open-source framework that can dynamically adapt to the changing needs of businesses, a notable shift from traditional ERP models. Their focus on the ZERP platform reveals its potential as a groundbreaking, intelligent ERP solution that harnesses blockchain's distributed ledger technology for enhanced flexibility, modularity, and scalability. The cloud-based SaaS architecture of ZERP is designed to be accessible to a spectrum of business sizes, from micro-enterprises to large corporations, supporting a diverse network of users and developers. This new generation of ERP systems aims to be highly collaborative and user-driven, allowing for continuous

improvement and evolution of the system, and laying a robust groundwork for a future where it could manage complex supply chains and serve as a comprehensive solution for various industries.

Banerjee [12], Shree et al. [13], Nageswar & Yellampalli [14], and Hader et al. [15] highlight the pivotal role of blockchain in transforming supply networks into decentralized, digital ecosystems. This transition, facilitated by blockchain, enhances both processes and products through the convergence of various technologies. The implementation of blockchain in supply chain processes promises enhanced visibility, transparency, and expedited decision-making. Moreover, blockchain technology is expected to lead to the creation of decentralized data repositories and the integration of analytics platforms, thereby boosting productivity, particularly in large-scale businesses. These studies also anticipate a future where systems like ERP and blockchain are highly interoperable, working in tandem to optimize operational efficiencies. The formation of blockchain-driven consortia is projected to transform the supply chain, making it more decentralized, interconnected, and efficient. This digital revolution, powered by blockchain, is not only beneficial for businesses but also has significant positive implications for society at large. The integration of blockchain into supply networks represents a significant leap forward in the evolution of supply chain management, marking a new era of digitalization and decentralization in business operations.

Extending the discussion from the transformative impact of blockchain in supply networks and business processes, Balasubramanyam [16] delves into its application in the manufacturing industry. The study highlights how the decentralized nature of blockchain technology can instigate significant changes in manufacturing processes, enhancing transparency, security, and operational efficiency. This aspect of decentralization is critical as it ensures data is not concentrated under a central authority, thus mitigating risks such as data manipulation and unauthorized access. Balasubramanyam [16] emphasizes that this feature of blockchain is not just a technical advantage but a fundamental shift in how manufacturing processes can be managed. By revolutionizing supply chain management, improving product traceability, and facilitating the use of smart contracts, blockchain's decentralization fosters a new level of trust and collaboration among industry stakeholders. This approach could lead to a comprehensive transformation in the manufacturing sector, redefining traditional practices and encouraging more secure, efficient, and transparent operations.

In essence, the intersection of blockchain technology with ERP systems, as delineated by Manceski & Nechkoska [17] and others, marks a critical shift towards decentralization, fundamentally redefining business process management. This paradigm shift brings decentralization to the forefront, disrupting traditional centralized models of ERP systems and extending its influence to supply chain management and manufacturing. The decentralizing power of blockchain not only enhances transparency and efficiency but also fosters a collaborative and inclusive business environment, as seen in the broader impacts discussed by Banerjee [12], Shree et al. [13], Nageswar & Yellampalli [14], and Hader et al. [15].

ERP & Supply Chain

ERP and blockchain technology are two powerful tools that can significantly enhance supply chain management. The supply chain represents the network of all the individuals, organizations, resources, activities, and technologies involved in the creation and sale of a product, from inception to delivery to the final consumer. Effective supply chain management is crucial to minimizing waste, reducing costs, and maximizing customer satisfaction, ensuring that the right goods reach the right place at the right time. When incorporated into an ERP system managing the supply chain, blockchain technology can significantly boost the efficiency, traceability, and security of operations. It offers a real-time, reliable, and verifiable record of each transaction, enhancing transparency, reducing fraud, and simplifying administrative tasks. This empowers businesses to optimize their supply chains, reduce costs and time delays, increase stakeholder trust, and ultimately deliver superior value to customers.

The integration of these two technologies has gained increasing attention due to its potential to revolutionize supply chain processes and contribute to more efficient, secure, and transparent operations. This section aims to explore the latest literature in this domain, discussing the current advancements, commonalities, differences, and areas of potential improvement identified by various scholars.

Sislian & Jaegler [18] discovered that incorporating blockchain into ERP modules can significantly improve supply chain strategies. This integration not only bolsters the sustainability of corporate performance but also strengthens supply chain integration. In their 2020 study, Liu et al. [19] explore the role of blockchain in revolutionizing logistics enterprises through strategic digital transformation. They emphasize the critical need for these enterprises to adapt to the rapidly evolving market dynamics. The authors propose a comprehensive framework that incorporates blockchain into digital logistics systems. This framework is designed to significantly enhance operational efficiency and bolster competitiveness, showcasing blockchain as a key tool in modernizing and optimizing logistics processes in the face of technological advancements.

Challenges in adopting blockchain technology are notable. Papathanasiou et al. [20] discuss the reluctance within the Greek shipping industry towards blockchain adoption, primarily due to concerns about the potential exposure of confidential business information, given blockchain's transparent nature.

On the other hand, some sectors are actively exploring the advantages of blockchain. For instance, Comuzzi's research in 2018 [21] delves into its implementation in the ERP system of a Mongolian livestock supply chain, underscoring substantial improvements in transparency and the reengineering of supply chain processes.

Further supporting blockchain's transformative potential, Lahlou and Motaki [22] and Hader et al. [15] discuss the revolutionary impact of blockchain on processes like procure-to-pay, emphasizing enhancements in transparency, data security, and supply chain management. They also recognize challenges such as scalability and infrastructure needs. Expanding on this, Faccia and Petratos [23] expand on these insights by suggesting that blockchain technologies can overcome the limitations of centralized ERP systems, notably in aspects of security and privacy. His research points to blockchain's capability to enable multi-level system integration, proving advantageous in fields such as auditing compliance, as demonstrated in e-procurement systems and operational analysis.

Expanding on the integration of blockchain and ERP for enhancing supply chain operations, Bhujade et al. [24] propose a dual-stage architecture that integrates on-chain and off-chain operations. This model is designed to improve the flexibility and efficiency of supply chain management. Despite its innovative approach, the model does face certain challenges. It notably enhances operational flexibility and reduces data storage costs. The research emphasizes the importance of achieving consensus among stakeholders and calls for a deeper investigation into the cost implications associated with data storage specific to ERP systems.

Complementing these insights, Riaz, Abrar, and Gul [25] add to the discussion on innovative technology solutions in supply chain processes. Although their work does not directly intersect with blockchain and ERP systems, it underscores the broader theme of integrating technology in supply chains. They developed a hybrid recommendation system that combines elements of both collaborative and content-based systems, demonstrating the potential of advanced technological integration in this sector.

In their targeted study, Lyakhovenko et al. [26] explore the implementation of a Smart Factory model that leverages smart contracts. This model is particularly relevant for resource management within supply chain operations, especially in manufacturing settings. It represents a concrete example of how innovative technologies, such as blockchain, can be integrated to increase the efficiency and effectiveness of supply chains. This research aligns with broader discussions in the field about harnessing cutting-edge technologies to optimize supply chain processes.

Recent literature in this domain reveals a consistent focus on blockchain's role in augmenting transparency, data security, and control within supply chains, particularly when merged with ERP systems. Researchers like Hader et al. [15] and Lahlou and Motaki [22] have proposed various methodologies to facilitate this integration, demonstrating its potential to revolutionize supply chain processes. Faccia and Petratos [23] highlights how blockchain can address the limitations inherent in centralized ERP systems, pointing towards a future where technology enables multi-level system integration, enhancing areas like auditing compliance.

The depth and approach of these studies vary. Hader et al. [15] and Lahlou and Motaki [22] offer broad perspectives on blockchain's potential applications in supply chain management. In contrast, other research focuses on more specific implementations. For instance, Faccia and Petratos [23] examine e-procurement systems, Lyakhovenko et al. [26] investigate a smart factory model leveraging smart contracts, and Bhujade et al. [24] propose a dual-stage architecture integrating on-chain and off-chain operations. Each study contributes unique insights into the benefits and challenges of integrating blockchain with ERP systems.

However, the adoption of blockchain is not without challenges. Papathanasiou et al. [20] discuss the hesitance within the Greek shipping industry towards blockchain, mainly due to concerns over the exposure of confidential business information.

The integration of blockchain technology with ERP systems holds significant potential to enhance supply chain management. While the literature reviewed reveals broad consensus on the potential benefits, such as improved transparency, data security, and process control, the methods proposed for integrating these technologies vary, each with its unique advantages and challenges. Continued exploration and innovation in this field is crucial to capitalize on these benefits and find the best methods to integrate blockchain with ERP systems, thus achieving more efficient, secure, and transparent supply chain management.

ERP & Accounting

The emergence of blockchain technology has generated considerable interest in its potential applications across diverse fields such as accounting and transaction management. This section is dedicated to exploring and synthesizing the key findings from nine research papers, each examining various facets of blockchain integration in areas like accounting information systems (AIS), and interfirm processes. Through an in-depth analysis of these papers, we aim to provide a comprehensive understanding of the challenges, opportunities, and future directions for implementing blockchain technology in these domains.

In their 2023 study, Alanazi and AlZubi [27] explore enhancing the efficiency and security of AIS transactions through blockchain technology. They introduce a novel algorithm designed to reduce the likelihood of blockchain splits, thereby ensuring more secure and efficient accounting transactions. This innovative approach is significant for optimizing transaction processes in AIS, leading to improved data integrity within these systems.

Also, the study by Gomaa et al. [28] complements the research by Alanazi and AlZubi [27] with its focus on blockchain technology in the context of AIS. This case study emphasizes the integration of blockchain with ERP systems and delves into the vital role of internal and external auditors in scrutinizing and validating transactions. Furthermore, it sheds light on the tax implications of cryptocurrency transactions and elaborates on the nexus between ERP systems and blockchain technology, thereby

broadening the understanding of blockchain's impact in financial and information systems.

The paper by Kitsantas and Chytis [29] focuses on the role of blockchain technology in the field of accounting and management. It offers insights into how blockchain can be effectively integrated into these fields, discussing the potential for blockchain to transform various aspects of accounting and management practices. The 2021 study by Sarwar et al. [30] examines the use of blockchain in enhancing the security and integrity of data in AIS. The study proposes a concept of data vaults backed by blockchain to ensure that financial and accounting data are tamper-proof and authentic. This approach aims to integrate blockchain into AIS and ERP systems, addressing challenges such as data mutability and integrity in traditional databases. The research demonstrates the potential of blockchain in providing robust solutions for secure data storage in accounting systems

Furthermore, Gomaa et al. [31] introduces an innovative single-ledger model leveraging blockchain technology in accounting. This model aims to eliminate fragmented information systems and reconcile multiple data versions, thereby streamlining accounting processes. It emphasizes enhancing efficiency by providing a unified and transparent ledger that can be accessed by multiple stakeholders, addressing the common reconciliation challenges in traditional accounting practices.

Additionally, in Zakharkina et al. [32], they explore how digital transformation, including the integration of ERP systems and blockchain, can enhance productivity, simplify transaction accounting, and streamline supply chain processes in the fuel and energy sector. The paper highlights the importance of continuous improvement in ERP systems and the establishment of a unified ecosystem for the sector.

Moreover, Lacity and Van Hoek [33] discuss the implementation of a blockchain-enabled invoice processing solution by Walmart Canada and its freight carriers. The study showcases the benefits of blockchain technology in reducing disputed invoices, improving transaction time, and enhancing cost savings. It provides recommendations for utilizing blockchain in interfirm process innovation and emphasizes the potential value of blockchains in improving interfirm processes.

Saxena and Pahwa [34] focus on improving the financial module of intelligent ERP systems by incorporating cryptocurrency, specifically Bitcoin. The paper discusses the growth of the cryptocurrency market, the different types of Bitcoins, and their utilization within the blockchain or ledger. It emphasizes the importance of using blockchain technology in financial transactions to ensure transparency, authenticity, and security.

Lastly, Jayasuriya and Sims [35] conducted a thorough review of academic and industry literature to examine the role of blockchain technology in accounting, comparing it with traditional legacy systems. Their paper identifies and discusses the current regulations, accounting standards, and guidelines that are relevant to blockchain adoption in the accounting sector. It also highlights the necessity for amendments in these regulations and standards to effectively integrate and leverage this emerging technology in accounting practices.

To maximize the potential of blockchain in accounting and ERP systems, organizations should explore the innovative methodologies and frameworks discussed in these research papers. These include optimized algorithms for improved transaction security, single-ledger models for unified data integrity, and transparent shared ledgers for enhanced visibility. The continuous evolution of ERP systems and the creation of cohesive ecosystems are vital for seamless blockchain integration. In essence, integrating blockchain with accounting and ERP systems promises a transformative impact on traditional practices, offering opportunities for more secure, efficient, and transparent financial operations

ERP & Security

The integration of blockchain technology in ERP systems offers transformative security enhancements. Primarily utilized in finance and cryptocurrency, blockchain's key features like immutability, consensus mechanisms, cryptographic hashing, and decentralized architecture can significantly bolster the security of ERP systems. These systems can leverage blockchain to ensure the integrity, confidentiality, and resilience of vital business data, deeply changing the approach to business data security.

While blockchain technology is prominent in finance and cryptocurrency, its application in ERP systems, as discussed by Gonçalves and Seruca [36], offers significant advantages. ERP systems integrate various corporate functions into one unified database and interface. By incorporating blockchain, ERP systems can become more secure, transparent, and efficient, unlocking new economic opportunities. Blockchain's immutability ensures that transactions or data entries within ERP systems are secure from unauthorized alterations, manipulation, and fraud, as highlighted by Gomez et al. [37]. This feature assures the accuracy and reliability of crucial business data within these systems.

Ben Othmane et al. [38] highlight that blockchain's consensus method allows only legitimate and authorized transactions to be recorded while preventing malicious actors from adding false or unauthorized transactions. Furthermore, blockchain leverages cryptographic hashing and encryption to enhance data security, ensuring that sensitive information remains confidential. Additionally, blockchain's distributed nature, as discussed by Kusnadi et al. [39], involves storing data across multiple network nodes, eliminating a single point of failure and enhancing the system's resistance to attacks or data loss. The redundancy and integrity of data across the network provide robust security measures, even in the event of a node compromise. Aslam et al. [40] explore the combination of Blockchain technology with ERP systems to enhance data protection and minimize the vulnerability to single-point failures. They propose employing a private Blockchain using Proof of Elapsed Time consensus to strengthen access management in ERP systems. Additionally, they tackle the

challenge of slow transaction processing in these systems. Their proposed approach involves a system design that integrates Blockchain and ERP using an application interface.

As per research conducted by Nageswar and Yellampalli [14], blockchain technology utilizes public-key cryptography and digital signatures to establish precise access control. This significantly reduces the risk of unauthorized access and ensures the secure and controlled utilization of sensitive data within an ERP system. The transparency inherent in blockchain technology enables the creation of a comprehensive audit trail for each transaction and its associated inputs. The transparent and traceable nature of changes made to the blockchain facilitates accountability and transparency. This, in turn, enhances security and compliance efforts by aiding in the detection and investigation of inconsistencies or anomalies within an ERP system, as emphasized by Chan et al. [41]. Furthermore, the literature suggests that blockchain-based smart contracts automate transactions based on predefined rules and criteria. By reducing the need for intermediaries and minimizing the potential for errors or malicious behavior by individuals, these smart contracts ensure precise and interference-free transaction execution. Consequently, they contribute to enhancing overall security, as discussed by Mann et al. [42].

Furthermore, research has indicated that traditional code analysis methods were historically employed to uncover software flaws. However, these methods have limitations when it comes to identifying logic errors, which serve as significant sources of vulnerabilities, including flawed authentication and access control mechanisms. To address this challenge, companies have implemented two distinct approaches for detecting code modifications that impact software security, as discussed by Gonçalves and Seruca [36]. In the first approach, senior engineers proactively tackle security issues before executing change requests. While effective, this method may entail a time-consuming process. Alternatively, the second method utilizes code annotations to capture developers' attention regarding security-related concerns, serving as alerts when they engage in code editing, as highlighted by Arora et al. [2]. While these methods have their merits, they predominantly focus on safeguarding the code itself and may fall short in providing comprehensive protection against fraud, data manipulation, and cyberattacks. This is where blockchain technology emerges as a valuable solution. Utilizing blockchain, an inherently secure distributed ledger, can enhance the security of ERP systems significantly. Blockchains offer several advantages, including immutability and resistance to manipulation. These features are achieved through a consensus mechanism that verifies transactions and data inputs, bolstered by cryptographic hashing for encryption. This ensures the integrity and reliability of critical corporate data. Moreover, the distributed and decentralized architecture of blockchain eliminates single points of failure, bolstering the system's resilience against intrusions and data loss. Another noteworthy aspect of blockchain technology is its transparency and auditability. This transparency allows for the creation of a comprehensive audit trail, simplifying the detection and investigation of anomalies, as emphasized by Ben Othmane et al. [38].

Further research has highlighted that the integration of blockchain into ERP systems plays a crucial role in helping businesses mitigate the risks of data manipulation, fraud, and cyberattacks, while also establishing a dependable source of truth and a verifiable audit trail [37]. Blockchain's transparency and immutability contribute to building stakeholder confidence, creating a secure environment for critical corporate processes. Additionally, blockchain streamlines workflows and processes by automating transactions and contracts, eliminating the need for intermediaries and paper-based procedures. This automation reduces the potential for human errors and data duplication, thus enhancing the system's safety and reliability. Furthermore, blockchain technology opens up new avenues for collaboration and value exchange through peer-to-peer networks, smart contracts, and security enhancements. It fosters loyalty, fuels innovation, sets products apart, and enhances consumer satisfaction, as highlighted by Nageswar and Yellampalli [14].

Among the gathered research, one study has discovered that blockchain technology enhances data security and privacy within the ERP system by implementing robust access control measures. This ensures that only authorized individuals or parties can access specific information. Blockchain achieves this by providing secure and selective data access through the utilization of encryption, public keys, and digital authentication, as highlighted by Chan et al. [41].

Mann et al. [42] suggest that blockchain technology holds significant potential for ensuring regulatory compliance within ERP systems. Its transparency and auditability facilitate the creation of verifiable audit trails, aiding in the demonstration of adherence to government and industry standards. By recording information and interactions on the blockchain, businesses can efficiently manage goods, processes, and supply chains, ensuring compliance with legislative requirements. The system can promptly issue alerts in case of deviations from compliance, enabling swift corrections. Furthermore, blockchain simplifies procedures by eliminating the need for paper-based testing and human tracking. Integrating blockchain with ERP systems provides instant access to information and automation, reducing reliance on physical paperwork and spreadsheets. This not only minimizes the risk of human error but also enhances overall data accuracy.

The integration of blockchain technology into ERP systems offers substantial security enhancements, with key features like immutability, consensus mechanisms, cryptographic hashing, and decentralization bolstering data integrity and confidentiality. Blockchain's advantages extend beyond its primary use in finance and cryptocurrency, as it enhances ERP system security, transparency, and efficiency. Immutability ensures the security of transactions and data entries, guarding against unauthorized alterations and fraud. Consensus mechanisms and

cryptographic techniques further fortify data security, while the decentralized architecture eliminates single points of failure and enhances resistance to attacks. Access control is improved through public-key cryptography and digital signatures, reducing the risk of unauthorized access. The transparency of blockchain facilitates audit trails, aiding in anomaly detection and compliance efforts. Smart contracts automate transactions, reducing errors and enhancing security. Despite these benefits, integrating blockchain can be complex and costly, scalability remains a concern, and regulatory uncertainty persists. User adoption, interoperability, security, and data privacy challenges must be carefully addressed. Nevertheless, blockchain's potential in enhancing ERP system security and compliance is promising.

ERP & Traceability

In various industries, particularly those involved in manufacturing and production, the role of traceability is critical. The integration of blockchain technology into ERP systems can significantly enhance this capability. Numerous research studies have explored how this integration can benefit and be applied across different industries. The convergence of blockchain with ERP systems can lead to more effective and efficient traceability, providing greater transparency, accuracy, and control over production processes and supply chains.

Numerous research articles have emphasized the critical importance of traceability in blockchain and ERP processes, especially in sectors like manufacturing and production. These studies explore various dimensions of traceability. For instance, Kusnadi et al. [39] focused on the traceability of halal food. This research underlined the necessity of reliable halal food standards, particularly when importing from countries where these standards might not be rigorously upheld. This poses a significant challenge for Muslim producers, necessitating a robust traceability system within the blockchain and ERP frameworks. The study highlighted how the synergy between blockchain and ERP systems could enhance the traceability of halal products. The proposed method involves recording primary halal food data into the ERP system, followed by the initiation of a contract that allows this information to be integrated into the blockchain network. This process ensures the validation of the data through various checks, thereby significantly reducing the risk of mixing halal with non-halal food. The integration of blockchain technology in this context not only ensures compliance with halal standards but also enhances the overall transparency and reliability of the supply chain.

Another study by Chan et al. [41] delved into the integration of blockchain technology and ERP systems, emphasizing their role in enhancing transparency and traceability in supply chain management, particularly within the agricultural food sector. This research specifically focused on how effectively blockchain technology can be implemented in supply chain management to improve traceability. The study revealed that certain characteristics of blockchain technology, particularly permissioned blockchains, are ideally suited for supply chain management applications. The incorporation of these blockchain types into supply chain processes could potentially lead to significantly improved transparency and traceability compared to other alternatives. This is attributed to the central control mechanism inherent in permissioned blockchains, which ensures high levels of data transparency and traceability. This aspect is especially crucial in the context of the Malaysian agricultural food sector, underscoring its vital role. Furthermore, Chan et al. [41] noted the existence of various forms of blockchains, both public and private, and stressed the importance of selecting the appropriate type for supply chain management. The choice is critical as different blockchain models have distinct algorithms, and any tampering or discrepancy could have significant implications. The study's insights highlight the nuanced considerations necessary in deploying blockchain technology for enhanced transparency and traceability in the agricultural food sector.

The final study in this section conducted by Mann et al. [42], highlights the vital importance of traceability and transparency in the mining industry, focusing on supplier relationships and commercial practices. It emphasizes the need for mining companies to be transparent and traceable, a demand driven by supplier platforms seeking honest and reliable information. Traditional methods for achieving this transparency have proven inadequate. Consequently, the study explores blockchain technology as a superior alternative. Blockchain serves as a secure digital ledger, recording transactions and maintaining data integrity without external interference, thus enhancing transparency and traceability. The study also notes the efficiency gains from blockchain, especially through the adoption of smart contracts, which improve process streamlining and accountability.

The studies covered in this section highlight the potential of blockchain technology and ERP systems for enhancing traceability procedures and highlight the significance of traceability in various industries. The study on the manufacture of halal food illustrated how traceability can help ensure the halal requirements, thereby addressing the difficulties faced by Muslim producers. Similarly, the study of transparent and traceable supply chain management in the agricultural food industry highlighted the advantages of utilizing blockchain technology to improve transparency and traceability.

ERP & Purchasing

ERP systems, when integrated with blockchain technology, can significantly enhance purchasing processes. This integration offers improved transparency, efficiency, and security, ultimately revolutionizing how purchasing operations are conducted. The adoption of blockchain in these systems not only streamlines the procurement process but also ensures a higher level of data integrity and trust in business transactions.

In the field of purchasing and ERP systems, numerous studies have investigated the benefits of incorporating blockchain technology to boost efficiency and

transparency. A notable example is the work by Shree et al. [13], which proposed a blockchain-integrated, smart, and secure purchasing system within an ERP framework. The study highlighted the use of the Internet of Things (IoT) for effective asset management, including an automated self-billing feature to streamline purchasing. This system, by harnessing the power of blockchain, provided unparalleled transparency across the supply chain, securely documenting product details at every step. Additionally, the integration of RFID (Radio Frequency Identification) technology for item tracking further enhanced security measures, particularly in preventing theft.

In their 2018 study, Tonnissen and Teuteberg [43], explored the application of blockchain technology in business processes related to purchasing. They identified the substantial advantages of integrating blockchain with ERP systems, particularly the use of smart contracts to boost efficiency and transparency in purchasing activities. Their research methodology included a case study, expert interviews, and qualitative content analysis, which provided insights into the challenges, barriers, and potential benefits of blockchain integration in purchasing operations

In 2021, Choudhari et al. [44] conducted a study that examined the management of subcontractor life cycles within enterprise systems using blockchain technology. While their primary investigation centered on subcontractor management, they proposed that this blockchain-based approach could also be beneficial for the purchasing processes in ERP systems. The research emphasized the critical need for enhanced visibility, trust, and real-time tracking of subcontractor information. By incorporating these subsystems into the blockchain network and utilizing smart contracts, the approach they suggested significantly improved transparency and mitigated revenue losses, leading to more efficient and effective purchasing processes.

The synthesis of blockchain technology with ERP systems marks a key advancement in the field of purchasing processes, promising a paradigm shift in operational efficiency, transparency, and control. This integration, as explored in recent studies (Shree et al. [13]; Tonnissen et al. [43]; Choudhari et al. [44]), brings forth a numerous of innovative applications such as IoT for asset management, automated self-billing systems, RFID-based item tracking, and smart contracts. These technologies converge to create a purchasing environment that is not only secure and automated but also deeply interconnected, ensuring a seamless flow of information and transactions. The promise of blockchain in ERP systems is transformative, offering a level of data integrity and trust in business transactions previously unattainable. However, the full potential of this integration remains to be completely harnessed, with ongoing research and practical implementations crucial for unlocking the full spectrum of benefits blockchain technology can bring to purchasing processes within ERP systems.

ERP & Process Automation

The integration of blockchain and process automation through smart contracts, as investigated by Kaid and Eljazzar [45], explores how blockchain can enhance traditional automation methods. Smart contracts offer businesses the ability to automate processes transparently and immutably across organizations without intermediaries, thus boosting stakeholder confidence. However, their adoption is hindered by challenges such as stakeholder mistrust, privacy concerns, and the inflexible nature of blockchain due to its immutable properties. This study underscores the significant potential of smart contracts in process automation, alongside the hurdles that need addressing for broader acceptance.

The research by Eggers et al. [46] delves into how blockchain technology, when integrated with ERP systems, positively impacts supply chain interactions. This integration facilitates enhanced data sharing, transaction processing, analytics, and business intelligence within supply chains. A key feature of this advancement is the automation of payments between distributors and retailers through the use of smart contracts on blockchain networks, which significantly improves the speed and security of transactions, thereby boosting the overall efficiency and performance of the supply chain.

Additionally, Scheer [47] discusses the role of blockchain technology in Business Process Management (BPM), which is aimed at restructuring business processes to align with digital transformation goals. This involves using various modeling techniques to identify and address inefficiencies in existing processes. The integration of blockchain, particularly smart contracts and Distributed Ledger Technology (DLT), has significantly influenced BPM by enhancing process automation capabilities, thereby improving the organization and execution of business processes in the digital era.

The research indicates significant potential for smart contracts and their integration with ERP systems in automating business processes. These studies also underline the necessity for further research to address implementation challenges, especially in complex and traditional business environments with legacy systems. In essence, while the union of blockchain, smart contracts, and ERP systems offers promising prospects for enhancing business efficiency, transparency, and performance, overcoming adoption and implementation hurdles remains key. Continued exploration in this field is essential to realize the full potential of these technologies and revolutionize traditional business operations.

ERP and blockchain integration

The integration of blockchain technology with ERP systems is a key focus in academic research, offering potential enhancements in data security, automation, and transparency. Sislian and Jaegler [18] explored the impact of this integration on sustainable corporate performance, finding significant improvements in supply chain strategies through their study involving structural equation modeling. This research underscores the transformative

potential of merging blockchain with ERP systems. Likewise, Liu et al. [19] explored the integration of artificial intelligence and blockchain technologies in digitally transforming logistics enterprises. Their research proposed a comprehensive digital logistics framework, emphasizing blockchain's vital role across various stages of digital transformation, from conceptual understanding to practical application. This study sheds light on the synergistic potential of combining AI and blockchain for enhancing logistics operations. Furthermore, Comuzzi [21] presented an innovative approach to system reengineering using blockchain technology. This research introduced a design space with real-world scenarios, demonstrating blockchain's applicability in tracking and managing changes in ERP systems. This study showcases the utility of blockchain in system reengineering, highlighting its potential to enhance process tracking and management. Fill and Härer [48] delved into the application of blockchain technology in enterprise modeling and introduced the innovative concept of "knowledge blockchains". This research not only emphasized but also reinforced the notion of utilizing blockchain technology as a novel paradigm for modeling and effectively managing knowledge within an organizational context.

Another prominent theme in the literature involves investigating the advantages that arise from integrating blockchain technology with ERP systems. Numerous benefits have been discerned, ranging from process automation and enhanced security to heightened transparency and improved traceability. Aslam et al. [40] examines integrating Blockchain technology with ERP systems to improve data security and reduce the risk of a single failure point. It suggests using a private Blockchain with Proof of Elapsed Time consensus for better access control in ERP systems and addresses the slow transaction processing issue. The proposed solution includes a system architecture that merges Blockchain with ERP through an application interface. Sislian and Jaegler [18], who illustrated that the integration of blockchain with ERP modules could significantly enhance supply chain strategies. This not only bolsters the sustainability of corporate performance but also strengthens the integration of supply chains. Similarly, the integration of blockchain in logistics offers significant advantages. Liu et al. [19] demonstrated how digital technologies, including blockchain, can enhance efficiency and competitiveness across various logistical processes. They highlighted blockchain's potential in preventing data tampering and improving transparency. In the financial sector.

Finally, Dewi et al. [49] explore the significant roles played by the integration of blockchain technology and the use of ERP systems in the development of infrastructure, particularly in the context of the emerging Industrial Revolution 5.0, which emphasizes the integration of technology with societal needs. They highlight how blockchain technology is focused on exploring its potential impacts and benefits, which contribute to the development of a more advanced and interconnected society. Furthermore, they point out the crucial role this integration plays in the successful deployment of ERP systems.

Although integrating blockchain with ERP systems offers a plethora of benefits, it is not without its challenges. Scholars have identified various technical and cultural obstacles that must be taken into account. Mougayar [7] argued that incorporating blockchain into existing systems, like ERP systems, can prove to be a challenging endeavor. These complexities arise from compatibility issues, the need for substantial modifications to the existing infrastructure, and the difficulty of establishing consensus mechanisms within a distributed network. Moreover, the absence of globally recognized legal standards and the concerns surrounding privacy in public blockchains present significant challenges for the integration of blockchain with ERP systems. The matter of transparency, often regarded as a blockchain advantage, can also pose challenges in specific industries. For example, Papathanasiou et al. [20] discovered that Greek shipping companies exhibited reluctance towards adopting blockchain technology because of its potential to expose sensitive business data. The perceived threat to competitive viability, resulting from the transparency of shared information, was considered a significant hurdle to the adoption of blockchain. Furthermore, the integration of blockchain with ERP systems typically demands a substantial allocation of resources and time. The hesitance to invest in innovation, especially within an uncertain economic environment, exacerbates these obstacles.

Blockchain technology's integration with ERP systems offers significant potential for improving various aspects of enterprise operations, from supply chain management to knowledge management and system reengineering. However, it is important to note that there are still considerable challenges to overcome, such as cultural resistance to change and the perceived threat to competitive survival due to the transparency of shared information. Future research should focus on further exploring the applications of blockchain in different sectors and addressing the challenges associated with its adoption.

CONCLUSION AND FUTURE DIRECTIONS

ERP systems, originating from material and manufacturing resource planning in the 70s and 80s, evolved significantly in the early 90s to integrate various business functions into a single software solution. Their evolution continued with the advent of the internet and cloud computing, leading to more adaptable, scalable, and efficient ERP architectures, further enhanced by data analytics and AI. Parallelly, blockchain technology, initially developed for Bitcoin, emerged as a decentralized ledger with potential applications in various sectors including healthcare and finance, promising to revolutionize aspects like transparency and security. The integration of blockchain and ERP systems, while promising for enhancing data integrity and process efficiency, is still an emerging field with limited practical applications. This paper aims to fill the research gap in this area, providing a comprehensive

analysis of the integration of blockchain into ERP systems, its current state, challenges, and future potential.

The research methodology employed in this study is a structured and systematic approach, divided into four key stages: Literature Retrieval, Literature Screening, Bibliometric Analysis, and Content Analysis. Initially, a comprehensive search using the Scopus database was conducted with specific keywords, resulting in a collection of 497 scholarly papers, which was narrowed down to 66 relevant publications after a rigorous screening process adhering to the PRISMA framework. The bibliometric analysis then examined citation patterns and co-authorship relations to identify key contributors and emerging trends in the integration of blockchain and ERP systems. Finally, in the content analysis phase, the selected publications were methodically reviewed and categorized to uncover prevailing patterns and insights into how blockchain technology impacts various aspects of ERP systems, thus providing a deeper understanding of their interplay and implications for enterprise management and strategy.

The bibliometric analysis of the integration of blockchain and ERP systems revealed several key findings. Despite the recognized demand for blockchain and ERP technologies across industries, actual implementations within ERP systems are scarce. The literature mainly focuses on their intersection, discussing benefits and challenges but lacking comprehensive insight into their current applications and future prospects. The analysis of 66 pertinent papers identified dominant terms like “Blockchain”, “ERP”, and “Cryptography”, indicating a strong link between blockchain technology and supply chain management, with an increasing focus on data security. The co-occurrence maps highlighted the emphasis on integrating blockchain with information systems, particularly in areas like smart contracts and data security. Geographic distribution analysis showed a majority of publications from Australia, India, and China, with rising interest in New Zealand and various EU countries. A diverse range of journals and sources explored the ERP-blockchain correlation, reflecting the broad interest and engagement in this research area. The findings underscore the evolving trends in blockchain's integration with ERP systems and the necessity for more focused research in this field.

The content analysis of the integration of blockchain technology with ERP systems revealed several key insights:

- ERP & decentralization: The study highlights the shift from centralized ERP systems to decentralized, blockchain-based frameworks. This shift offers enhanced flexibility, adaptability, and inclusiveness in ERP solutions. Platforms like ZERP exemplify this transition, emphasizing open-source, cloud-based, and modular architecture that caters to various business scales, promoting a dynamic, decentralized approach in ERP system development.
- ERP & supply chain: Research underscores the significant potential of integrating blockchain with ERP systems in supply chain management. Blockchain's role in enhancing transparency, traceability, and efficiency in supply chains is emphasized, with studies showing how it can revolutionize logistics, reduce fraud, and improve sustainability and integration of corporate performance.
- ERP & accounting: Investigations into the role of blockchain in accounting reveal its potential to enhance transaction security, streamline processes, and improve data integrity. Innovations like single-ledger models and transparent shared ledgers are discussed, illustrating how blockchain can transform traditional accounting practices and integrate seamlessly with ERP systems.
- ERP & security: The integration of blockchain in ERP systems significantly bolsters security, leveraging features like immutability, cryptographic hashing, and decentralized architecture. This integration is shown to enhance data integrity and confidentiality, offer robust access control measures, and facilitate regulatory compliance.
- ERP & traceability: Studies highlight the critical role of blockchain in enhancing traceability in various industries. Blockchain's integration with ERP systems is shown to ensure higher transparency and control over production processes and supply chains, especially in sectors like halal food production, agriculture, and mining.
- ERP & purchasing: Blockchain's integration with ERP systems in purchasing processes promises enhanced transparency, efficiency, and security. The use of IoT, smart contracts, and RFID technology in blockchain-integrated ERP systems is discussed, demonstrating how it can transform purchasing operations.
- ERP & process automation: The role of blockchain in automating business processes through smart contracts is explored. Challenges in adoption are acknowledged, but the potential for enhancing efficiency, transparency, and stakeholder confidence in various business operations is highlighted.
- ERP and blockchain integration: The content analysis acknowledges the challenges in integrating blockchain with ERP systems, including technical, cultural, and infrastructural barriers, and the need for substantial resource allocation. Despite these challenges, the integration promises substantial improvements in various enterprise operations, necessitating further research to address these obstacles.

The integration of blockchain technology into ERP systems is fraught with several formidable challenges. Technical complexities and integration issues present significant barriers, as existing ERP systems often require substantial modifications to align with blockchain's compatibility requirements. Additionally, blockchain technology's scalability limitations pose threats to the performance of integrated systems. Cultural and organizational resistance to change, especially in sectors with well-established ERP frameworks, further impedes blockchain's adoption. Privacy and security concerns also emerge due to blockchain's inherent transparency, which can be problematic in industries that manage sensitive data. The absence of global legal standards for blockchain complicates

regulatory and compliance adherence, particularly for international operations. Lastly, the resource intensiveness of adopting and implementing blockchain within ERP systems constitutes a substantial hurdle, especially for smaller organizations that may lack the necessary financial and human capital.

The future of research in blockchain and ERP systems integration is expected to take a comprehensive approach, addressing scalability to better fit the integration with ERP systems of varying sizes and complexities. This involves the development of new blockchain algorithms and architectures to facilitate scalability and performance enhancements. Seamless interoperability between blockchain and a multitude of ERP platforms, including the adaptation of legacy systems, is also a critical area of focus, ensuring smooth integration and data exchange. Specifically, research will likely develop tailored blockchain integration models for various industries such as healthcare, finance, and manufacturing, each with its distinct challenges and opportunities. Innovations in privacy-preserving mechanisms within blockchain, employing advanced cryptographic techniques like zero-knowledge proofs, will aim to maintain transparency while enhancing transaction privacy. The amalgamation of AI and blockchain is anticipated to transform ERP systems into intelligent platforms capable of automated decision-making and predictive analytics. Economic impact analyses are necessary to clarify the financial implications and potential returns of blockchain integration across different business environments. Blockchain as a Service (BaaS) offerings are expected to be refined to cater to ERP systems, prioritizing ease of adoption, flexibility, and cost-effectiveness, which could be particularly beneficial for small and medium-sized enterprises (SMEs). Research into effective change management strategies and cultural adaptation is vital for facilitating the organizational acceptance of blockchain technology. Standardization efforts and the development of a cohesive regulatory framework will guide the legal and compliance aspects of blockchain and ERP integration, addressing cross-border transactional challenges. Additionally, a focus on the environmental impact of blockchain technology will be essential, leading to the development of sustainable and energy-efficient blockchain solutions. Enhancing supply chain traceability is another prospective research avenue, using blockchain within ERP systems to track products from source to consumer, particularly in critical sectors such as agriculture and pharmaceuticals. Real-time data analytics integrated into blockchain-ERP systems will be explored to offer businesses timely insights for informed decision-making. Lastly, developing robust security protocols and frameworks will be crucial for safeguarding blockchain-integrated ERP systems from cyber threats and vulnerabilities, ensuring the integrity and resilience of business data.

REFERENCES

[1] E.M. Shehab *et al.* "Enterprise resource planning: An integrative review," *Business Process Management Journal,* vol. 10, *(4),* pp. 359-386, 2004. DOI: 10.1108/14637150410548056.

[2] R. Arora, S. Gera and M. Saxena, "Mitigating security risks on privacy of sensitive data used in cloud-based ERP applications," in *2021 8th International Conference on Computing for Sustainable Global Development (INDIACom),* 2021.

[3] S. Fosso Wamba *et al.* "How 'big data' can make big impact: Findings from a systematic review and a longitudinal case study," *Int J Prod Econ,* vol. 165, pp. 234-246, 2015. Available: DOI: 10.1016/j.ijpe.2014.12.031.

[4] L. Ghiro *et al.* "What is a Blockchain? A Definition to Clarify the Role of the Blockchain in the Internet of Things," *arXiv Preprint arXiv:2102.03750,* 2021.

[5] S. Nakamoto, "Bitcoin: A peer-to-peer electronic cash system," *Decentralized Business Review,* 2008.

[6] D. Tapscott and A. Tapscott, *Blockchain Revolution: How the Technology Behind Bitcoin is Changing Money, Business, and the World.* 2016.

[7] W. Mougayar, *The Business Blockchain: Promise, Practice, and Application of the Next Internet Technology.* 2016.

[8] S. Saberi *et al.* "Blockchain technology and its relationships to sustainable supply chain management," *Int J Prod Res,* vol. 57, *(7),* pp. 2117-2135, 2019. DOI: 10.1080/00207543.2018.1533261.

[9] M.M. Queiroz, R. Telles and S.H. Bonilla, "Blockchain and supply chain management integration: a systematic review of the literature," *Supply Chain Management: An International Journal,* vol. 25, *(2),* pp. 241-254, 2020. DOI: 10.1108/SCM-03-2018-0143.

[10] R. Casado-Vara *et al.* "How blockchain improves the supply chain: case study alimentary supply chain," *Procedia Computer Science,* vol. 134, pp. 393-398, 2018. DOI: 10.1016/j.procs.2018.07.193.

[11] N. Kshetri, "1 Blockchain's roles in meeting key supply chain management objectives," *Int. J. Inf. Manage.,* vol. 39, pp. 80-89, 2018. DOI: 10.1016/j.ijinfomgt.2017.12.005.

[12] A. Banerjee, "An ERP and planning system enabled decentralized supply chain using blockchain technology," in *Blockchain Driven Supply Chains and Enterprise Information Systems* Anonymous 2022.

[13] J. Shree *et al.* "To design smart and secure purchasing system integrated with ERP using block chain technology," in *2020 IEEE 5th International Conference on Computing Communication and Automation (ICCCA)*Anonymous 2020, Available: https://aus.on.worldcat.org/oclc/8696648713. DOI: 10.1109/ICCCA49541.2020.9250767.

[14] A.K. Nageswar and S. Yellampalli, "Distributed trust using blockchain for efficient pharmaceutical supply chain," in *Global Supply Chains in the Pharmaceutical Industry* Anonymous 2019.

[15] M. Hader, A. El Mhamedi and A. Abouabdellah, "Blockchain integrated ERP for a better supply chain management," in *Proceedings of the 2021 8th International Conference on Industrial Engineering and Applications (Europe),* 2021.

[16] K. Balasubramanyam, "Adaptation of block chain technology in manufacturing sector," in *2020 International Conference on Mainstreaming Block Chain Implementation (ICOMBI),* 2020.

[17] G. Manceski and R. Petrevska Nechkoska, "Conceptualisation of decentralized blockchain-based, open-source ERP marketplaces: Disruptive decentralized technologies for co-creation," in *Facilitation in Complexity: From Creation to Co-Creation, from Dreaming to Co-*

*Dreaming, from Evolution to Co-Evolution*Anonymous 2023.

[18] L. Sislian and A. Jaegler, "Linkage of blockchain to enterprise resource planning systems for improving sustainable performance," *Business Strategy and the Environment,* vol. 31, *(3),* pp. 737-750, 2022.

[19] H. Liu *et al*, "Strategy-oriented digital transformation of logistics enterprises: The roles of artificial intelligence and blockchain," in *2020 5th International Conference on Innovative Technologies in Intelligent Systems and Industrial Applications (CITISIA),* 2020.

[20] A. Papathanasiou, R. Cole and P. Murray, "The (non-) application of blockchain technology in the Greek shipping industry," *European Management Journal,* vol. 38, *(6),* pp. 927-938, 2020.

[21] M. Comuzzi, E. Unurjargal and C. Lim, "Towards a design space for blockchain-based system reengineering," in *Advanced Information Systems Engineering Workshops: CAiSE 2018 International Workshops, Tallinn, Estonia, June 11-15, 2018, Proceedings 30,* 2018.

[22] I. Lahlou and N. Motaki, "Integrating blockchain with ERP systems for better supply chain performance," in *2022 14th International Colloquium of Logistics and Supply Chain Management (LOGISTIQUA),* 2022.

[23] A. Faccia and P. Petratos, "Blockchain, enterprise resource planning (ERP) and accounting information systems (AIS): Research on e-procurement and system integration," *Applied Sciences,* vol. 11, *(15),* pp. 6792, 2021.

[24] V. Bhujade *et al.* "Perpetual interoperability of legacy ERP and blockchain in supply chain," in *2021 5th International Conference on Information Systems and Computer Networks (ISCON),* 2021.

[25] R. Riaz, S. Abrar and M.J. Gul, "Securing supply-chain network with blockchain," in *Future of Information and Communication Conference,* 2023.

[26] J. Lyakhovenko, I.I. Viksnin and S. Chuprov, "Integrating smart contracts into smart factory elements' informational interaction model." in *Micsecs,* 2020.

[27] J. Mutared Alanazi and A. Ali AlZubi, "An Optimized Method for Information System Transactions Based on Blockchain," *Intelligent Automation & Soft Computing,* vol. 35, *(2),* pp. 2289-2308, 2023. DOI: 10.32604/iasc.2023.029181.

[28] A.A. Gomaa, M.I. Gomaa and A. Stampone, "A Transaction on the Blockchain: An AIS Perspective, Intro Case to Explain Transactions on the ERP and the Role of the Internal and External Auditor," *Journal of Emerging Technologies in Accounting,* vol. 16, *(1),* pp. 47, 2019. DOI: 10.2308/jeta-52412.

[29] T. Kitsantas and E. Chytis, "Blockchain Technology as an Ecosystem: Trends and Perspectives in Accounting and Management," *Journal of Theoretical and Applied Electronic Commerce Research,* vol. 17, *(3),* pp. 1143, 2022. DOI: 10.3390/jtaer17030058.

[30] M.I. Sarwar *et al.* "Data vaults for blockchain-empowered accounting information systems," *IEEE Access,* vol. 9, pp. 117306-117324, 2021.

[31] A.A. Gomaa *et al*, "The Creation of One Truth: Single-Ledger Entries for Multiple Stakeholders Using Blockchain Technology to Address the Reconciliation Problem," *Journal of Emerging Technologies in Accounting,* vol. 20, *(1),* pp. 59, 2023. DOI: 10.2308/JETA-19-06-01-28.

[32] L. Zakharkina *et al.* "The Impact of Digital Transformation in the Accounting System of Fuel and Energy Complex Enterprises (International Experience)," *International Journal of Energy Economics and Policy,* vol. 12, *(5),* pp. 152-161, 2022. DOI: 10.32479/ijeep.13317.

[33] M.C. Lacity and R. Van Hoek, "How Walmart Canada Used Blockchain Technology to Reimagine Freight Invoice Processing," *MIS Quarterly Executive,* pp. 219-233, 2021. DOI: 10.17705/2msqe.00050.

[34] K. Saxena, A.S. Pahwa and International conference on Computer Networks, Big data and IoT Madurai, India December 19, 2019 - December 20, 2019 ICCBI, "Enhanced financial module in intelligent ERP using cryptocurrency," in *Proceeding of the International Conference on Computer Networks, Big Data and IoT (ICCBI - 2019)*Anonymous 2020, Available: https://aus.on.worldcat.org/oclc/8549870494. DOI: 10.1007/978-3-030-43192-1_106.

[35] D.D. Jayasuriya and A. Sims, "From the abacus to enterprise resource planning: is blockchain the next big accounting tool?" *Accounting, Auditing & Accountability Journal,* vol. 36, *(1),* pp. 24-62, 2023 DOI: 10.1108/AAAJ-08-2020-4718.

[36] D. Gonçalves and I. Seruca, "SEPA Files Transmission: Implementing security guarantees in enterprise resource planning systems," 2018.

[37] L. Gomez *et al.* "Security for distributed deep neural networks towards data confidentiality & intellectual property protection," 2019.

[38] L.B. Othmane, A. Jamil and M. Abdelkhalek, "Identification of the impacts of code changes on the security of software," in *2019 IEEE 43rd Annual Computer Software and Applications Conference (COMPSAC),* 2019.

[39] A. Kusnadi *et al.* "Designing Halal Product Traceability System using UML and Integration of Blockchain with ERP," vol. 9, *(1),* pp. 29-41, 2023.

[40] T. Aslam *et al.* "Blockchain based enhanced ERP transaction integrity architecture and PoET consensus," vol. 70, *(1),* pp. 1089-1109, 2022.

[41] K.Y. Chan, J. Abdullah and A.S. Khan, "A framework for traceable and transparent supply chain management for agri-food sector in malaysia using blockchain technology," vol. 10, *(11),* 2019.

[42] S. Mann *et al.* "Blockchain technology for supply chain traceability, transparency and data provenance," in *Proceedings of the 2018 International Conference on Blockchain Technology and Application,* 2018.

[43] S. Tonnissen, F. Teuteberg and 21st International Conference on Business Information Systems, BIS 2018 21st 2018 07 18 – 2018, 07 20, "Using blockchain technology for business processes in purchasing – Concept and case study-based evidence," vol. 320, pp. 253-264, 2018. DOI: 10.1007/978-3-319-93931-5_18.

[44] S. Choudhari *et al.* "Sub contractor life cycle management in enterprise system using blockchain technology," in *2021 6th International Conference for Convergence in Technology (I2CT)* Anonymous 2021, Available: https://aus.on.worldcat.org/oclc/9033584119. DOI: 10.1109/I2CT51068.2021.9417981.

[45] D. Kaid and M.M. Eljazzar, "Applying blockchain to automate installments payment between supply chain parties," in *2018 14th International Computer Engineering Conference (ICENCO),* 2018, .

[46] J. Eggers *et al*, "Process automation on the blockchain: an exploratory case study on smart contracts," 2021.

[47] A. Scheer, "The development lines of process automation," in *The Art of Structuring: Bridging the Gap between Information Systems Research and Practice*, K.

Bergener, M. Räckers and A. Stein, Eds. 2019, DOI: 10.1007/978-3-030-06234-7_20.
[48] H. Fill and F. Härer, "Knowledge blockchains: Applying blockchain technologies to enterprise modeling," 2018.
[49] V. Dewi, A. Amelia and T.N. Mursitama, "Blockchain technologies and factor of ERP implementation in making infrastucture for society 5.0," in *2021 3rd International Conference on Cybernetics and Intelligent System (ICORIS),* 2021.

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